



TrintzERP

RETAIL & BILLING PLATFORM

User & Technical Guide

A complete walkthrough of every feature — point-of-sale billing, GST compliance, inventory, credit & supplier management, reporting, and the built-in AI assistant — written for business owners and IT teams alike.

GST & e-Invoice (IRN)

Inventory

Receivables & Payables

Tally Export

AI Assistant

Role-based Access

Cloud Backup

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Introduction & Overview

TrintzERP is a complete, web-based Point-of-Sale and retail management platform built for Indian businesses. It runs in any modern web browser and combines fast counter billing with full GST compliance, inventory control, supplier and customer credit tracking, accounting-grade reporting, Tally integration, and an built-in AI assistant — all in a single, role-controlled application.

Who it is for



Retail Shops

Fast billing, barcode-friendly product search, printed receipts, day-close reporting.



Distributors

Purchase orders, supplier payables, stock valuation and bulk data import.



GST Businesses

GST-compliant invoices, e-Invoice (IRN), GSTR-style reports and Tally export.

What makes it different



All-in-one

Billing, inventory, credit, GST, reports and admin in one platform — no separate tools to reconcile.



AI-assisted

Ask business questions in plain English; the assistant answers safely from your own data.



Role-secure

Four roles and a fine-grained, screen-level permission system protect every function.



Backed up

Scheduled local and Google Drive backups keep your data safe and recoverable.

How to read this guide

Sections 1–16 are written for everyday business users (owners, managers, cashiers). Section 17 onward is for IT/technical staff and covers architecture, security and deployment. A How-To quick reference and FAQ are at the end.

Getting Started & Login

Accessing the application

Open a modern web browser (Google Chrome or Microsoft Edge recommended) and go to the address provided for your installation — for an on-site setup this is typically `http://localhost:5000`, or the URL/IP given by your administrator.

Logging in

1. Enter your **username** and **password** on the login screen.
2. If **Two-Factor Authentication (2FA)** is enabled for your account, enter the 6-digit code from your authenticator app (e.g. Google Authenticator).
3. On success you are taken to the Dashboard (or to billing, depending on your role).

First-time setup

Default accounts (such as an initial administrator) must have their passwords changed immediately after first login. Your administrator will provide your personal credentials.

License activation

TrintzERP is a licensed product. After login the system automatically validates your license. If a valid license is not found, you are guided to the **License Activation** screen — enter the license key provided with your purchase to unlock full access. Once activated you are returned to the Dashboard automatically.

Tip — staying signed in securely

Always use **Logout** when leaving the counter, especially on shared machines. Sessions are protected by secure tokens and will expire for safety.

User Roles & Permissions

TrintzERP uses **role-based access control** backed by a fine-grained, screen-level permission system. Every user is assigned one role; administrators can then further allow or restrict individual screens per user.

Role	Typical user	Access
Super Admin	Business owner / system owner	Full, unrestricted access to every module including administration, permissions, backups and AI settings.
Admin	Store manager / senior staff	Broad operational access (sales, purchases, inventory, reports, customer/supplier credit, cancellations) subject to granted permissions.
Manager	Shift / department lead	Operational access to assigned areas; typically reporting and day-to-day management functions.
Cashier	Counter staff	Focused on billing and sales; restricted from administrative and financial-control screens.

Screen-level permissions

Beyond the role, Super Admins can grant or revoke access to **individual screens** (e.g. Reports, Token Usage, Credit Management) for each user from the **Administration → Access Permissions** panel. The system follows a **deny-by-default** policy: a user only sees a screen if their role or an explicit permission allows it.

How it protects you

Access is enforced on the server, not just hidden in the menu. Even a direct attempt to open a restricted page is blocked and redirected to an **Access Denied** screen.

The Dashboard

The Dashboard is your business at a glance, shown immediately after login. It summarises the day's performance and highlights items that need attention.



Today & Month Sales

Total sales amount, invoice count and GST collected for today, yesterday and the current month, with day-over-day change indicators.



Low Stock Alerts

A live list of products that have fallen below their reorder level, so you can restock before running out.



Top Products

Best-selling items by quantity and value to guide purchasing decisions.



Outstanding Credit

Total receivables — money customers owe you — surfaced directly on the dashboard.

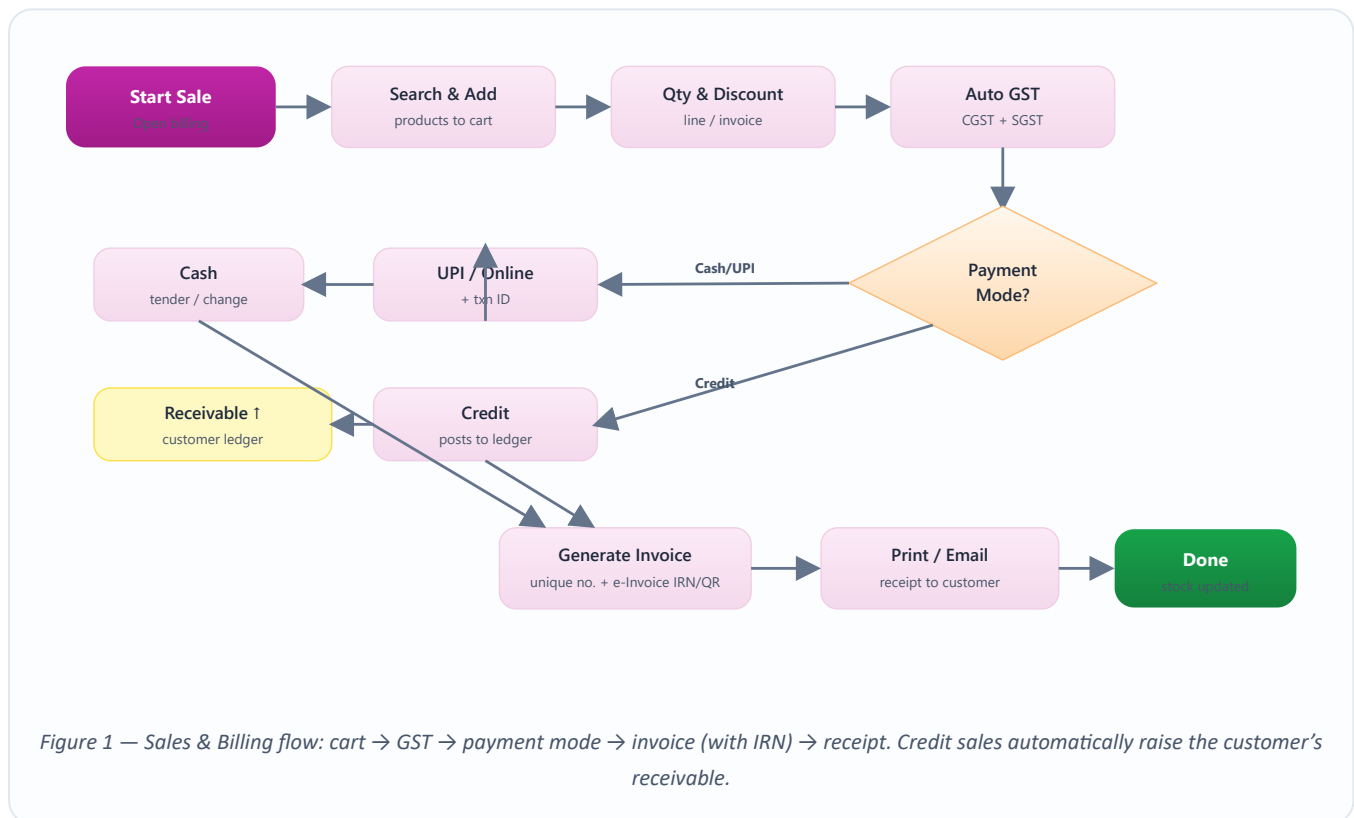
Recent transactions are listed for quick reference, and the left navigation menu adapts to show only the modules your role and permissions allow.

Sales & Billing

The Sales screen is the heart of TrintzERP — designed for fast, accurate counter billing with automatic GST calculation and multiple payment modes.

Creating a sale — step by step

1. **Find products:** search by name or SKU and add them to the cart.
2. **Set quantity & discount:** adjust quantity and apply line or invoice-level discounts.
3. **GST is automatic:** CGST/SGST is calculated per product from its GST rate.
4. **Add customer (optional):** capture name and mobile for receipts and credit.
5. **Choose payment mode:** Cash, UPI/Online (with transaction ID) or Credit.
6. **Complete sale:** the invoice is generated with a unique number; an e-Invoice IRN and QR are attached automatically when enabled.
7. **Print/share receipt:** print or email the receipt to the customer.



Credit sales & the ledger

When a sale is billed as **Credit**, TrintzERP automatically records the amount against the matched customer's account (a Tally-style receivable). Cancelling such an invoice reverses the entry automatically, keeping your books in sync.

Stock safety

The system checks stock availability before completing a sale and decrements inventory as items are sold, preventing overselling.

Products & Inventory

Product Management

Maintain your full catalogue: product name, SKU, pack size, HSN code (for GST), selling rate, GST rate, and reorder level. Products can be added, edited and removed by authorised users.

Field	Purpose
Name / SKU	Identify and quickly search the product at billing.
HSN Code	Required for GST-compliant invoicing and reports.
Selling Rate	Default price used at billing (editable per sale if allowed).
GST Rate	Drives automatic CGST/SGST calculation.
Reorder Level	Triggers low-stock alerts on the dashboard.

Inventory Management

Track live stock levels for every product. Stock decreases automatically on sales and increases on purchases/stock receipts. Manual stock adjustments are available to authorised users for corrections and stocktakes.

Low-stock workflow

Products below their reorder level appear in the Dashboard's Low Stock list, prompting a purchase order before you run out.

Purchases & Suppliers

Purchase orders & stock receiving

Record incoming stock from suppliers. Each purchase captures items, quantities, purchase rates and GST, and automatically updates inventory levels. Purchase order numbers are generated for traceability.

Supplier Management

Maintain a supplier master with GST number, contact and banking details. Supplier balances track what your business owes (payables).

Concept	Meaning (Tally convention)
Supplier balance > 0	Payable — your business owes the supplier.
Supplier balance < 0	Advance Paid — you have paid in advance.
“Credit” transaction	Adds to payable (e.g. a credit purchase).
“Debit” transaction	Payment made to the supplier (reduces payable).

Credit Management — Receivables & Payables

TrintzERP provides a unified, accounting-grade view of money owed to you (customer receivables) and money owed by you (supplier payables), following standard Indian (Tally) conventions.

 **Customer Receivables**

Track per-customer outstanding balances. A **credit sale / charge** increases what a customer owes you; a **payment received** reduces it. View statements, ageing and transaction history per customer.

 **Supplier Payables**

Track what you owe each supplier, record payments, and see advances. Outstanding amounts roll up into the credit dashboard.

The Credit Dashboard

A single overview shows total **Receivables (To Collect)**, total **Payables (To Pay)**, and your **Net Position**. Customer advances are reported separately so they are never confused with money owed to you.

Term	Plain meaning
Receivable	Money a customer still has to pay you.
Payable	Money your business still has to pay a supplier.
Advance	A customer has paid more than due / you paid a supplier ahead.
Net Position	Total receivables minus total payables.

Always in sync

Credit sales, payments, returns and cancellations all post consistently to the ledger, and an automatic reconciliation flags any account whose stored balance drifts from its transaction history.

Ageing & statements

Outstanding amounts are bucketed by age (how overdue they are) using a first-in-first-out settlement model, and per-account statements can be exported to Excel for follow-up.

GST Reports & Tally Export

GST compliance

Every invoice is GST-aware: CGST and SGST are computed per item from the product's GST rate and HSN code. The GST Reports module summarises outward tax (on sales), input tax credit (ITC on purchases), and the **net tax payable** for a chosen period — the figures you need for return filing.

e-Invoice (IRN & QR)

When enabled, qualifying sales automatically receive an **Invoice Reference Number (IRN)** and a signed **QR code**, attached to the invoice for statutory e-invoicing.

Tally export

Export your sales, purchases and returns as Tally-compatible vouchers (XML). The export uses correct double-entry: credit sales debit the customer (Sundry Debtor); purchases credit the supplier (Sundry Creditor); returns post as Credit Notes — so your accountant can import directly into Tally.

Accountant-friendly

GST summaries and Tally export are designed so your CA or accounts team can reconcile and file without re-keying data.

Returns & Invoice Cancellation

Sales returns

Process customer returns against an original invoice. Returns restore stock, adjust GST correctly, and record the refund method — and they flow into reports and Tally export as Credit Notes.

Invoice cancellation

Authorised users (typically Admin/Super Admin) can cancel an invoice. On cancellation the system:

- Restores the sold stock back to inventory;
- Marks the invoice as Cancelled with an audit trail (who, when, reason);
- **Automatically reverses** any credit-ledger entry created by that sale, restoring the customer's receivable.

Controlled action

Cancellation is permission-controlled and fully audited to prevent misuse.

Reports & Analytics

TrintzERP includes a reporting suite to understand performance and profitability.



Sales Reports

Sales by period, with totals, GST and trends; visual charts for quick reading.



Purchase Reports

Purchases by period and supplier, for spend analysis and reconciliation.



Profit & Loss

Revenue versus cost of goods to reveal gross margins and profitability.



Inventory Reports

Stock valuation and movement to support stocktakes and planning.

Reports can be filtered by date range and exported for sharing with management or your accountant.

AI Assistant

TrintzERP includes a built-in **AI Assistant** that lets you ask questions about your business in plain English — for example, “Which customers owe us the most?” or “What were today’s total sales?” — and receive instant, data-driven answers.

How it works (safely)

Your question is sent to the AI, which translates it into a **read-only** database query, runs it against your own data, and returns a clear answer. The assistant is restricted to safe, read-only lookups — it cannot modify, delete or expose protected information.

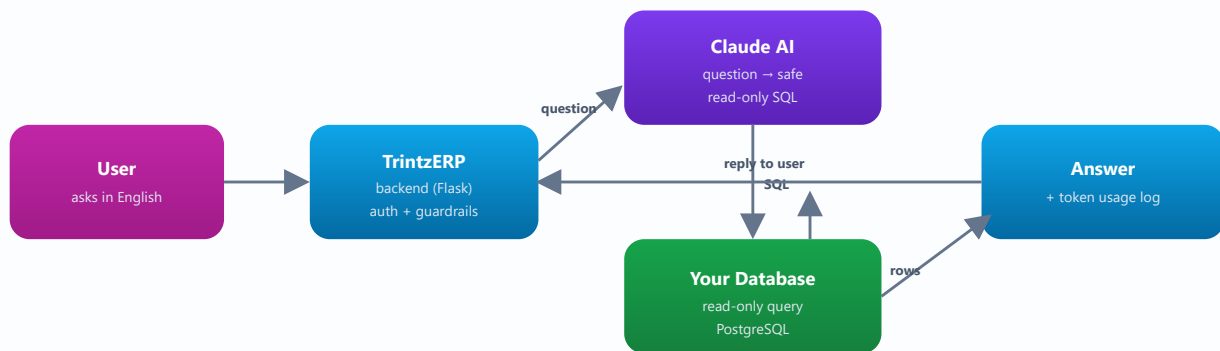


Figure 2 — AI Assistant: questions are converted to safe, read-only queries over your own data; usage is metered.

AI Settings & Token Usage

Administrators configure the AI from **AI Settings** and monitor consumption from the **Token Usage** screen, which tracks how much AI capacity has been used. Access to these screens is permission-controlled.

Knowledge Base

A built-in Knowledge Base lets you store reference articles and help content that the assistant and your staff can draw on.

Data Upload & OCR

Bulk Excel upload

Save hours of data entry by importing from Excel. Downloadable templates ensure correct formatting.

Products

Bulk add/update products; existing SKUs are updated automatically.

Sales

Import historical sales with GST integrity preserved.

Purchases

Import purchase history; inventory and suppliers updated automatically.

OCR / document import

The OCR-to-Excel tool extracts tabular data from documents (e.g. supplier bills/PDFs) to speed up data entry and reduce manual typing errors.

Validate before importing

Always use the provided template and review the preview; the system reports rows it could not process so you can correct them.

Administration

The Admin area (Super Admin / Admin) centralises control of the system.



User Management

Create users, assign roles, reset passwords and activate/deactivate accounts.



Access Permissions

Grant or revoke individual screens per user with a clear, categorised permission grid.



Login Activity

Review successful and failed login attempts for security monitoring.



Invoice Viewer

Search, view and audit all invoices across the business.



Store Settings

Configure business details (name, GST number, address) used on invoices and reports.



Email Invoice

Email invoices/receipts to customers directly from the system.

Admin Panel — Every Tile Explained

The Admin Panel is the single entry point for all administrative tools, grouped into three sections: **Management**, **Data Tools** and **Configuration**. A live stats bar at the top shows Products, Suppliers and Users counts.

How tile visibility works

Each tile appears only if the user **both** (1) meets the tile’s minimum role rank — **L2 Manager+** **L3 Admin+** **L4 Super Admin only** — and (2) has the tile’s screen permission granted. This is deny-by-default and enforced on the server, not merely hidden from the menu.

Management

Day-to-day control of master data and activity monitoring.

Tile	What it does	Min role
 Products	Add, edit & delete products — name, SKU, HSN, selling rate, GST rate, reorder level (catalogue master).	Manager+
 Suppliers	Maintain supplier master (GST number, contact, bank details) and view each supplier’s ledger/payables.	Manager+
 Users	Create staff accounts, assign roles, reset passwords, activate/deactivate.	Admin+
 Invoice Viewer	Browse, search and reprint every invoice; audit past sales.	Manager+
 Credit Accounts	Customer credit management — receivables, balances, statements, payments.	Manager+
 Login Activity	Monitor successful and failed logins for security oversight.	Manager+
 Token Usage	Track AI consumption per user and model.	Manager+

Data Tools

Import, export and database utilities — moving data in and out of the system.

Tile	What it does	Min role
 Data Upload	Bulk-import products, sales & purchases from Excel templates.	Admin+
 OCR → Excel	Extract tabular data from images/PDFs (e.g. supplier bills) into Excel.	Manager+
 Query to DB	Run custom SQL queries against the database — a power-user/diagnostics tool.	Admin+
 Tally Export	Export sales, purchases & returns as Tally-compatible XML vouchers.	Manager+
 Auto Backup	Schedule and manage database backups (local + Google Drive), and restore.	Admin+

Query to DB is the most sensitive tool here

It allows direct SQL access. It is gated to Admin+ and should be tightly restricted.

Configuration

System-wide setup: business identity, access control, AI and licensing.

Tile	What it does	Min role
 Store Settings	Inline panel to set business name, address, phone, GSTIN and UPI ID (with QR). Used on invoices/receipts.	Manager+
 Permissions	Opens the Access Permissions panel — grant/revoke individual screens per user.	Super Admin only
 Service	Database health & schema tools — DB status and maintenance utilities.	Manager+
 AI Settings	Configure the LLM provider & API keys powering the AI assistant.	Admin+
 Knowledge Base	Manage RAG documents — reference content the AI draws on for answers.	Admin+
 License	View and manage the software license (activation status, key).	Admin+

Permissions is the only Super Admin-exclusive tile

Because it controls who can access everything else, it is deliberately the most restricted. Store Settings and Permissions open inline panels within the Admin Panel; all other tiles navigate to their own page.

Backup & Data Safety

Protecting your business data is built in. TrintzERP supports both local and cloud backups.



Local Backups

Create on-demand or scheduled database backups stored safely on the machine.



Google Drive Backup

Connect a Google account to automatically copy backups to the cloud for off-site safety.



Scheduled

Backups can run automatically on a schedule so you never forget.



Restore

Restore from a backup to recover data after hardware failure or error.

Recommended practice

Enable scheduled backups *and* Google Drive sync, and periodically test a restore so you know your recovery works before you ever need it.

Security Features

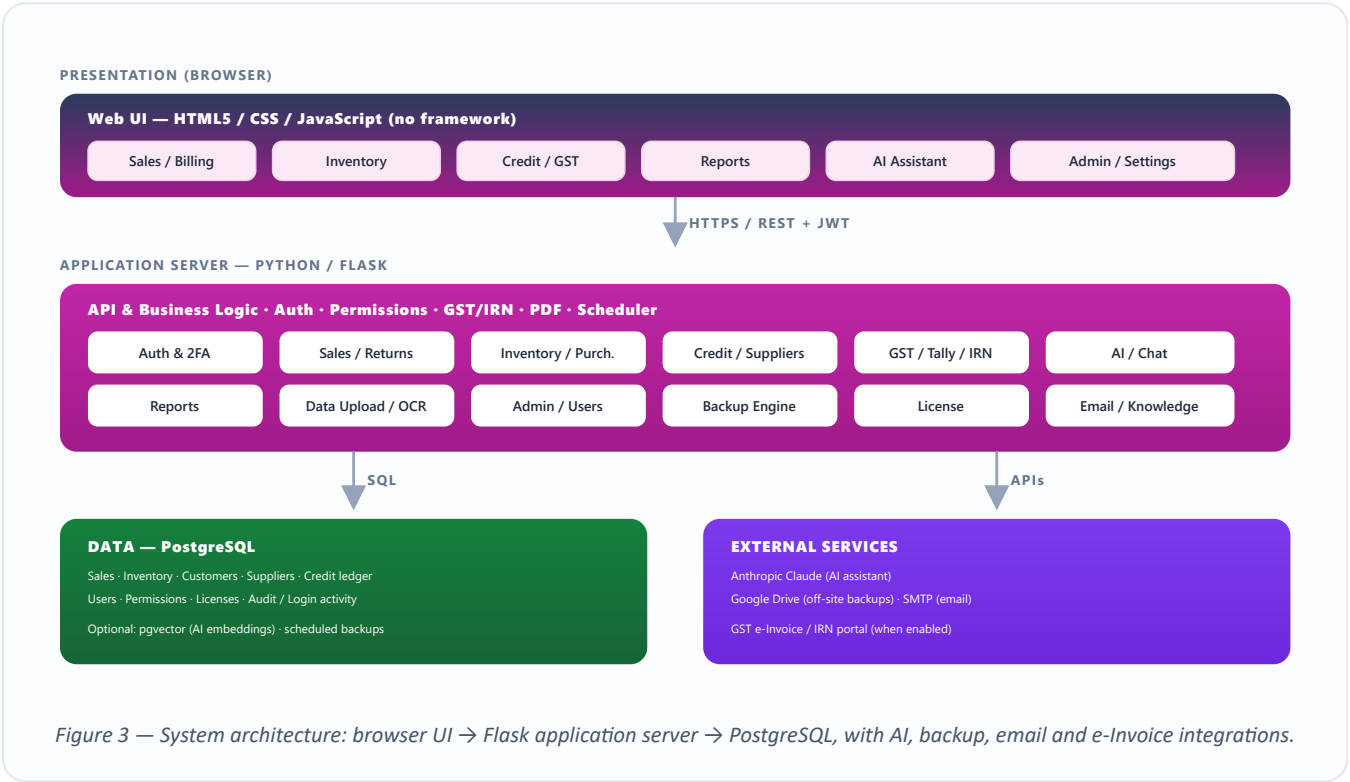
Feature	What it protects
Encrypted passwords	Passwords are stored using strong one-way hashing — never in plain text.
Secure session tokens (JWT)	Each session is authenticated by a signed token that expires.
Two-Factor Authentication (2FA)	Optional time-based one-time codes add a second layer at login.
Role & screen permissions	Deny-by-default access control, enforced on the server.
Login activity log	Visibility into successful and failed login attempts.
Audit trails	Sensitive actions (e.g. cancellations) record who, when and why.
Rate limiting & security headers	Protects against abuse and common web attacks.
License activation	Ensures only licensed installations operate.

Defence in depth

Authorisation is enforced by the backend on every request — restricted screens and data remain protected even against direct access attempts, not merely hidden from menus.

Technical Architecture

This section is intended for IT and technical staff. TrintzERP is a three-tier web application: a browser-based front end, a Python/Flask application server, and a PostgreSQL database, integrated with a small number of external services.



Technology stack

Layer	Technology
Front end	HTML5, CSS3, vanilla JavaScript; Chart.js for visualisation
Application server	Python 3 + Flask (Waitress/Gunicorn in production)
Database	PostgreSQL (with optional pgvector for AI features)
Authentication	JWT session tokens; passlib password hashing; PyOTP 2FA
AI	Anthropic Claude via the official SDK
Documents	openpyxl / xlswriter (Excel), xhtml2pdf (PDF), PyMuPDF (OCR/PDF)
Integrations	Google Drive API (backup), SMTP (email), GST e-Invoice (IRN)
Scheduling	APScheduler (automated backups & jobs)

Deployment options

- **On-premise (single shop):** installed on a counter PC/server with a local PostgreSQL; accessed at `http://localhost:5000` or over the shop LAN.
- **Cloud / hosted:** deployed to a cloud host (e.g. Render) and accessed via a URL.

How-To Quick Reference

I want to...	Do this
Bill a customer	Sales → search products → add to cart → set qty/discount → choose payment → Complete → print receipt.
Sell on credit	Add the customer (name + mobile) → choose <i>Credit</i> as payment mode → complete. The receivable updates automatically.
Add a new product	Product Management → Add Product → fill name, SKU, HSN, rate, GST, reorder level → Save.
Receive new stock	Purchase → create a purchase entry → add items & quantities → Save (inventory updates automatically).
Record a customer payment	Credit Management → open the customer → add a <i>Payment received</i> transaction.
Pay a supplier	Supplier Management → open supplier → add a <i>Debit (Payment Made)</i> transaction.
Process a return	Returns → select original invoice → choose items → set refund method → Save.
Cancel an invoice	Invoice Viewer / Sales → open invoice → Cancel → enter reason (Admin only; stock & ledger reverse automatically).
See who owes me money	Credit Management / Credit Dashboard → Receivables (To Collect).
File GST	GST Reports → select period → review outward tax, ITC and net payable → export.
Export to Tally	Tally Export → select period & document types → download the XML → import into Tally.
Import products in bulk	Data Upload → download the product template → fill it → upload → review results.
Ask the AI a question	Open the AI Assistant → type your question in plain English → read the answer.
Add a user	Administration → User Management → Add User → set role & password.
Restrict a screen for a user	Administration → Access Permissions → select user → toggle the screen off.
Back up my data	Backup → Create Backup now, or enable scheduled + Google Drive backup.
Enable 2FA	Profile/Security (TOTP) → scan the QR with an authenticator app → confirm the code.

Frequently Asked Questions

Do I need an internet connection to use TrintzERP?

For an on-premise installation, day-to-day billing and management work on your local machine/LAN without internet. Internet is needed only for features that use external services — the AI assistant, Google Drive backup, email and GST e-Invoice (IRN) generation.

Which web browser should I use?

Any modern browser works; Google Chrome or Microsoft Edge are recommended for the best experience. The interface is also responsive for tablets.

How is GST calculated?

Each product carries a GST rate and HSN code. At billing, the system automatically splits tax into CGST and SGST per item, and the GST Reports module rolls these up for your return period.

What does a negative customer balance mean?

Following Tally convention, a negative balance means the customer *owes you* money (a receivable). A positive balance means the customer is in advance/credit. The screens label these clearly as “Receivable” and “Advance”.

What happens to stock and credit when I cancel an invoice?

Cancellation restores the sold stock to inventory and automatically reverses any credit ledger entry created by that sale, so both inventory and receivables stay correct. The action is audited.

Is the AI assistant safe with my data?

Yes. The assistant is limited to safe, read-only queries against your own database and cannot change or delete data. Administrators can configure it and monitor usage via Token Usage.

Can multiple staff use it at once?

Yes. Each user logs in with their own account and role; permissions control what each person can see and do.

How do I recover if the computer fails?

Restore from your most recent backup. With scheduled local + Google Drive backups enabled, an off-site copy is available. Test a restore periodically so you're confident in recovery.

Can my accountant work with this?

Yes — GST summaries export for filing, and sales/purchases/returns export as Tally-compatible vouchers for direct import.

I forgot my password — what do I do?

Ask a Super Admin/Admin to reset it from User Management. For security, only administrators can reset passwords.

Why can't I see a particular menu/screen?

Access is permission-controlled. If you need a screen that isn't visible, ask your administrator to grant it from Access Permissions.

Glossary & Support

Term	Meaning
SKU	Stock Keeping Unit — a unique code identifying a product.
HSN	Harmonized System of Nomenclature code — classifies goods for GST.
CGST / SGST	Central / State GST — the two halves of intra-state GST.
ITC	Input Tax Credit — GST paid on purchases, offset against GST on sales.
IRN	Invoice Reference Number — the unique ID for a registered e-Invoice.
Receivable	Money customers owe your business.
Payable	Money your business owes suppliers.
Net Position	Total receivables minus total payables.
2FA / TOTP	Two-Factor Authentication using time-based one-time codes.
JWT	A signed token that securely identifies a logged-in session.

Support

For assistance, configuration help, or to report an issue, contact your TrintzERP support representative. When reporting a problem, please include your username, the screen you were on, what you expected, and what happened — this helps resolve issues quickly.

Thank you for choosing TrintzERP.

This guide covers the platform's core capabilities. Features and screens may be tailored to your specific installation and the permissions assigned to your account.